Object Oriented Programming (CSC166)
Lab Sheet 2

1. Write a program that defines a class "Student" with private data members for roll number, name, and marks obtained in 5 subjects. Use a parameterized constructor to initialize the roll number and name, and a member function to input the marks. Compute the total, percentage, and grade, then display the complete result.

Hint: [*getMarks() → inputs marks for 5 subjects*
calculateResult() → computes total, percentage, and grade
Percentage ≥ 90 → Grade A, ≥ 75 → B, ≥ 60 → C, ≥ 40 → D, < 40 → F
displayResult() → prints roll number, name, marks, total, percentage, and grade]

2. Write a program that defines a class "Employee" with private data members empId, name, basicSalary, hra, da, grossSalary, tax, and netSalary. Use a parameterized constructor to initialize empId, name, and basicSalary, and member functions to calculate the gross salary and net salary.

Hint: *[calculateGrossSalary() → grossSalary = basicSalary + hra + da*
hra = 20% of basicSalary, da = 10 % of basicSalary
calculateNetSalary() → tax = 10% of grossSalary if grossSalary >50000
netSalary = grossSalary − tax
displaySalarySlip() → prints a neatly formatted salary slip for the employee]

3. Write a program that defines a class "Room" with private data members length, breadth, and height (in feet). Implement a default constructor, a parameterized constructor, and a copy constructor. Write member functions to compute the area of the floor and the volume of the room, and a function to display the room's dimensions along with the computed area and volume.

Hint: *[Room() → default constructor, initializes all dimensions to 10*
Room(double l, double b, double h) → parameterized constructor
Room(const Room &r) → copy constructor performing a deep copy of the dimensions
calculateArea() → returns length × breadth
calculateVolume() → returns length × breadth × height]

4. Write a program that defines a class "ComplexNumber" with private data members real and imag. Implement a default constructor and a parameterized constructor. Write member functions to add and subtract two complex numbers, where one object is passed as a function argument and a new object is returned as the result. Display the complex numbers in the form a + bi.

Hint: *[ComplexNumber addComplex(ComplexNumber c2) → returns sum as a new object*
ComplexNumber subComplex(ComplexNumber c2) → returns difference as a new object
displayComplex() → prints the number in the form a + bi (or a − bi if imag is negative)
Test in main() by creating two ComplexNumber objects and displaying their sum and difference]

5. Write a program that defines a class "Counter" with a private static data member count that keeps track of the total number of objects created. Use a constructor that increments count every time an object is created, and a destructor that decrements count every time an object is destroyed. Implement a static member function to return the current value of count.

Hint: [*static int count; → declared inside the class, defined and initialized to 0 outside the class*
Counter() → constructor increments count and displays a message
~Counter() → destructor decrements count and displays a message
static int getCount() → static member function that returns count
Demonstrate in main() by creating several objects in different scopes and printing getCount() after each creation and destruction, showing that count is shared across all objects]